

Response to the Phase 0 AWS Setup, Cost Estimate & Execution Plan

LocalAI TV · Architecture Verdict + Founder Status Review · 18 May 2026

To: Sheikh Sameer & Gyana Rajnan (AI Pipeline Engineers) · cc Nagarjuna Reddy (CTO) **For the Founder:** Koneti Mohan Reddy (Managing Director) **From:** LocalAI TV Architecture Team **Re:** (1) verdict on the AWS Setup/Cost Plan, (2) GPU-quota readiness, (3) the Founder's 5 questions, (4) what is still missing before Zero Testing and Phase 1 **Status:** 🟡 **Gate remains HELD.** The plan is sound and approved *in design*. Execution still waits on the GPU quota approval, the still-open items, and the Founder's explicit gate-lift after re-verification.

TL;DR — plain language for the Founder

The AWS plan from Sameer & Gnana is good, careful, and technically correct. They took the safer option at every fork:

- The earlier dangerous DNS idea (handing the whole domain to Cloudflare, which could have broken your live Play Store app) is **gone**. They are now using **AWS CloudFront with a single Hostinger CNAME** — your live app on `aiservices.localaitv.com` is **not touched**. This was the one critical risk in the last round; it is now fully defused.
- The database move (Google Cloud → AWS, PostgreSQL 16) and the storage/CDN design are correct, with a 7-day rollback safety net.
- The cost is small and realistic: **~\$20/month** for storage + CDN + database. The GPU machine is the real cost and is correctly tracked separately and **gated**.

On the GPU quota (your headline concern): the team requested **8 vCPU for "Running On-Demand G and VT instances" in Mumbai**. This is **exactly the right request** — it is precisely what our runbook (item A0) told them to ask for. It is enough for Zero Testing. The only problem is **AWS has not approved it yet** — that is an AWS-side wait, not a mistake by the team. Escalating the support case is the correct action; there is nothing more to "fix" on our side for the quota itself.

Where we actually are: we are at the **pre-Zero-Testing** stage. We are **not** close to Phase 1 yet — and that is correct and safe. Your sequence (Zero Testing → Internal Validation → Controlled Rollout → Phase 1) is the right one. Today we are still *before* step 1, waiting on (a) the AWS quota approval and (b) the remaining sign-off items. **Good news:** almost all of the setup work (database, storage, CDN) can be done *in parallel while waiting for AWS* — so the wait is not wasted time.

This does not lift the gate. Nothing is executed until you say "start," after the Architecture Team re-verifies the completed work.

1. Verdict on the AWS Setup & Cost Plan — Endorsed in design (with minor refinements)

The document correctly implements all three Architecture-Team decisions from the 18 May DB/CDN verdict:

Item	Plan's choice	Verdict
B-1 — CDN	AWS CloudFront + S3, <code>content.localaitv.com</code> as a Hostinger CNAME only , nameservers untouched, private bucket + OAC	 Now FULLY endorsed. The critical DNS production-risk is defused. They also took the <i>stronger</i> security option (private bucket + Origin Access Control) instead of public-read — better than the minimum we asked for.
B-2 — Database	Migrate <code>newsai_db</code> GCP Cloud SQL → AWS RDS PostgreSQL 16 , <code>ap-south-1</code> , private SG, 7-day GCP read-only fallback	 Endorsed. Co-located, disciplined cutover, tested rollback.
B-3 — Version	PostgreSQL 16 + <code>pg_cron</code> via custom parameter group + reboot	 Endorsed. Conservative, correct.

Technical claims I checked and confirm are accurate:

- ACM certificate **must be in** `us-east-1` for CloudFront — correct and non-negotiable; good that they called it out.
- **S3 → CloudFront origin transfer is free**, and the **CloudFront 1 TB / 10M-request perpetual free tier** — both accurate as of current AWS pricing.
- **S3 Block Public Access ON + OAC, CloudFront principal only** — this is the correct secure pattern and supersedes the earlier public-read idea. Approved.
- Single-AZ RDS for Phase 0, Multi-AZ before Phase 1 — correct staging.
- Region lock `ap-south-1` (ACM the only `us-east-1` exception) — correct.
- ~\$20/month estimate (excl. EC2 GPU) — realistic and consistent with official pricing.

Minor refinements (NOT blockers — apply during execution, record in the completion report):

1. **RDS size:** start Phase 0 on `db.t4g.small` (**2 GB RAM**), not `t4g.micro` (1 GB). Restoring `newsai_db` plus `pgvector` + `pg_cron` on 1 GB risks an out-of-memory failure *during validation itself*. The cost difference (~\$13/month) is trivial against the risk of a stalled gate. Keep `micro` only if the restored dataset is confirmed tiny.
2. **GPU instance must not run 24x7 in Phase 0.** The g4dn is the dominant cost. For Zero Testing, **start it for a test run, then STOP it** between runs. Left running 24x7, a g4dn.xlarge in Mumbai is roughly ₹30–35k/month; start/stop keeps Phase 0 GPU cost to a fraction of that. Make this an explicit runbook line.
3. **Add earlier billing alerts.** The single alert at \$364.70 is fine as a ceiling, but add **\$50 and \$150 tiers** plus an AWS Budgets monthly budget, so an overage is caught early, not near the cap.
4. **Confirm the instance type is** `g4dn.xlarge` (4 vCPU, 1× T4, NVENC-capable) for Phase 0 — cheapest NVENC option and fits the 8-vCPU quota with headroom (see §2).

2. GPU Quota Readiness — the Founder's headline question · Request is correct; approval is the only blocker

The request — "8 vCPU, Running On-Demand G and VT instances, `ap-south-1` (Mumbai)" — is **exactly correct**. It matches `PHASE-0-RUNBOOK` item A0 word-for-word ("request at least 8 to allow one running + one test instance"). Specifically, 8 vCPU allows:

- one `g4dn.xlarge` (4 vCPU, 1× NVIDIA T4) running **plus a second 4-vCPU test instance**, or
- one `g4dn.2xlarge` (8 vCPU, 1× T4).

Either is **more than enough for Zero Testing**, which is single-stream NVENC transcode validation (gate G1: bulletin build < 2 min vs ~15 min baseline). **No sizing change is needed for Phase 0.**

Honest caveats — please note these now so there are no surprises later:

1. **This 8-vCPU quota is a Phase-0 quota, not a Phase-1 quota.** Phase 1 = many constituencies / concurrent channels = many GPU vCPUs. A **much larger quota increase will be needed before Phase 1**, and AWS GPU-quota approvals have **long, unpredictable lead times**. Recommendation: the moment Zero Testing passes, **immediately file the Phase-1 quota request** (with the projected channel count) so the approval runs in the background during Internal Validation/Controlled Rollout instead of blocking the launch.
2. **Brand-new AWS accounts default to 0 here**, so this approval *genuinely* blocks Part B execution. The team escalating Case 177892985400464 is the correct and only action — there is no engineering fix on our side; it is an AWS support timeline.
3. **"On-Demand" quota only.** If Spot instances are used later for cost savings, that is a *separate* quota ("All G and VT Spot Instance Requests"). Not needed for Phase 0; flag it for Phase-1 cost planning.
4. **The quota wait does not block anything else.** RDS provisioning, the GCP→RDS migration rehearsal, S3 bucket creation, CloudFront + ACM + OAC, IAM, DNS CNAME staging — **none of these need the GPU**. They should all proceed in parallel now (as pre-gate prep), so that the day the quota lands, only the GPU steps remain.

Verdict: GPU-quota readiness is **correct and on the right track**. Status = *waiting on AWS*, not *blocked by an error*.

3. The Founder's 5 Questions — direct answers

Q1. What additional requirements are needed from your side (Founder)?

- **Decision authority only at the end:** you lift the gate **after** the Architecture Team re-verifies the completed work — not before.
- **Approvals the team is waiting on you/CTO for:** region lock (`ap-south-1`) confirmed final; RDS size confirmed (we recommend `t4g.small`); OAC private-bucket pattern confirmed; a **maintenance window** approved for the GCP→RDS cutover (live channels currently use GCP).
- **CTO action (Nagarjuna):** route all credentials **only through the Bitwarden vault** — never to the AI team in chat, never to the Architecture Team, never by screen-share/AnyDesk. (This rule is firm and unchanged.)
- **Access for verification:** give the Architecture Team read access to the **Admin Dashboard repo** so the Supabase-Mumbai host can be positively confirmed (currently an assumption).

Q2. What technical updates must the AI team complete before Zero Testing?

Close every item in the plan's §9 plus the runbook's Part A:

1. **Admin DB verified** from the Admin repo (Supabase Mumbai — confirmed, not assumed).
2. **S-2:** S3 bucket `localaitv-content-mumbai` created in `ap-south-1`.
3. **S-1:** bucket-scoped IAM policy drafted to replace `AmazonS3FullAccess` (apply before Phase 1; document now).
4. **C1–C4 answered + Gnana & Abishek sign-offs** (RTMP fan-out phase, Phase-1 CloudFront cost, ABR ladder vs GPU budget, pg_boss at scale).
5. **Pre-gate build done in parallel:** RDS provisioned + pg_cron parameter group, S3 + CloudFront + ACM + OAC configured, GCP→RDS migration **rehearsed** (dump/restore + row-count/checksum) — *staged, not cut over*.
6. **GPU quota approval** (Case 177892985400464) confirmed.

Note: completing 1–6 is *prep*. Actually running the pipeline on the GPU (Part B / G1–G4) still only happens **after the Founder lifts the gate**.

Q3. Are there missing infrastructure / deployment / streaming / API / scalability preparations?

Yes — and this is important to state honestly. This plan is excellent for what it covers (storage + CDN + database), but it deliberately does **not** cover several things that are needed before Phase 1:

- **Live streaming architecture is not yet specified.** This document is the **VOD / processed-media** path (pipeline → S3 → CloudFront). **Live multi-channel streaming and RTMP fan-out (C1) is still an open question**, not a designed system. Zero Testing only validates single-stream NVENC encoding — it does **not** validate live fan-out.
- **The API layer is out of scope here.** `aiservices.localaitv.com` (your live app's API) has no scaling/capacity plan in this doc. It is correctly left untouched for Phase 0, but a Phase-1 API scaling + RDS-connection plan is still owed.
- **Monitoring & logging is a real gap.** The plan has *billing* alerts only — there is **no CloudWatch metrics/alerts/dashboard and no log aggregation**. You cannot properly judge G1–G4 or catch a pipeline failure without this. A minimal observability setup must be added to Zero Testing (see §5).
- **Admin Dashboard backend is not wired.** Today it is an **in-app demo with no backend**. The real Admin ↔ Supabase ↔ pipeline-DB federation is unbuilt and untested — a separate Phase-1 workstream.
- **Load and security testing is not planned.** The plan has good security *checks* (IAM/MFA, private SG, Block Public Access, OAC) — but no test that *proves* the DB is unreachable from the internet, the bucket is not publicly listable, and the CDN serves only intended paths; and no load test of the pipeline/CDN. Lightweight for Phase 0; required before Phase 1.
- **Phase-1 DB HA / right-sizing** (Multi-AZ, read replica, instance class under real load) — correctly deferred, but on the Phase-1 list.

Q4. What validations/testing must be completed before Phase 1?

Map onto your own sequence:

- **Zero Testing — gates G1–G4 must all PASS:**
 - **G1 Encoding latency:** a bulletin builds in **< 2 min** (target ~90s) vs the ~15-min baseline, using GPU NVENC.
 - **G2 S3 reliability:** 100% of test builds land at the correct `s3://localaitv-content-mumbai/ai-processed/...` key.
 - **G3 CDN delivery:** the same media is fetchable via `https://content.localaitv.com/ai-processed/...` (through the CDN, not the S3-direct URL).
 - **G4 End-to-end:** media is playable through the CDN within the build cycle with **no manual steps**.
- **Internal Validation:** post-cutover **live-channel check** (the 9 channels still work after the DB move), DNS/CDN cache-hit proof, RDS smoke test, and a **tested rollback drill** (repoint to GCP).
- **Controlled Rollout:** enable for **one constituency / small %** first, with monitoring live, before widening.
- **Then Phase 1**, only after the above pass and you say "start."

Q5. Recommendations by area

Area	Recommendation
AWS infrastructure	Plan endorsed. Apply the §1 refinements: <code>t4g.small</code> for Phase 0, start/stop the GPU, tiered billing alerts, confirm <code>g4dn.xlarge</code> . Do all non-GPU prep now in parallel.
GPU quota readiness	8 vCPU request is correct for Phase 0. Keep the case escalated. Pre-file the larger Phase-1 quota the moment Zero Testing passes (long AWS lead time).
Streaming architecture	Out of scope of this doc and not yet designed for live . Resolve C1 (RTMP fan-out) and the live ingest→transcode→package→deliver path as a separate Phase-1 design before any production streaming. Zero Testing validates VOD encoding only.
Database stability	Endorsed. <code>t4g.small</code> Phase 0; Multi-AZ + PITR + a tested restore drill before Phase 1 ; keep GCP read-only 7 days; final snapshot before decommission.
Front-end & admin testing	App is being reviewed page-by-page (separate, on track). Admin Dashboard real backend is unbuilt — schedule Admin↔Supabase↔pipeline-DB integration + testing as an explicit Phase-1 workstream.
Security & load testing	Security <i>design</i> is good. Add active tests : confirm RDS not internet-reachable, bucket not public/listable, CDN path-scoped; add a controlled pipeline/CDN load test before Phase 1.
Monitoring & logging	Add now (Zero-Testing gap): CloudWatch alarms for RDS CPU/memory/connections, EC2 GPU utilization, S3 4xx/5xx, CloudFront error-rate + cache-hit ratio, and pipeline job success/fail count + duration; ship pipeline logs to CloudWatch Logs. Without this, G1–G4 results are not trustworthy and failures are invisible.

4. Where We Actually Are — status vs. the Founder's roadmap

Founder's sequence: Zero Testing → Internal Validation → Controlled Rollout → Phase 1

WE ARE HERE _____▲ (pre-Zero-Testing)

Blocked by: (a) AWS GPU quota approval [waiting on AWS]
(b) Gate HELD pending §9 open items + re-verification

Not blocked: all non-GPU setup (RDS / S3 / CloudFront / ACM / OAC / migration rehearsal)

— do this in parallel NOW so launch day is GPU-only.

We are **not near Phase 1** — and that is the correct, safe position. The plan reflects this honestly; please do not be pressured (by timelines or by anyone) into lifting the gate before G1–G4 pass and the Architecture Team re-verifies.

5. Next Steps

- 1. **AI team:** keep Case 177892985400464 escalated; do **all non-GPU prep in parallel** (RDS `t4g.small` + pg_cron group, S3 bucket, CloudFront + ACM + OAC, GCP→RDS migration rehearsal, tiered billing alerts, minimal CloudWatch alarms + log shipping).
- 2. **AI team:** close §9 — verify Admin DB host, draft S-1 scoped policy, confirm S-2 bucket, answer C1–C4, collect Gnana + Abishek sign-offs.
- 3. **CTO (Nagarjuna):** technical sign-off; route credentials **via the vault only**; grant Architecture Team read access to the Admin repo.
- 4. **AI team → Architecture Team:** on quota approval + §9 closure, submit the **completion report** (G1–G4 capture, post-cutover live-channel check, DNS/CDN proof, rollback-drill result) for **re-verification**.
- 5. **Founder:** lift the Phase 0 gate **only after** re-verification passes. Then Part B / Phase 1 proceeds per `PHASE-0-RUNBOOK`.
- 6. **Founder/Architecture (forward-looking):** treat **live streaming architecture (C1), the API scaling plan, the Admin backend, and load/security testing** as explicit Phase-1 design workstreams — they are *not* covered by this Phase-0 document.

Sign-off

Party	Action	Status
Sameer & Gnana	Non-GPU prep in parallel; close §9; keep quota escalated	☐ Pending
Gyana Rajnan	DB migration plan final sign-off	☐ Pending
Abishek	C1–C4 architecture sign-off	☐ Pending
Nagarjuna (CTO)	Technical sign-off; vault-only credentials; Admin repo access	☐ Pending
Architecture Team	Re-verify on completion report (G1–G4 + cutover + DNS + rollback)	☐ Awaiting
Founder — Koneti Mohan Reddy	Lift gate only after re-verification	☒ Holding

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Gate held. AWS Setup/Cost Plan endorsed in design; B-1 now fully endorsed (DNS risk defused). 8-vCPU GPU quota request is correct for Phase 0; approval is an AWS-side wait. v1.3 unchanged; CloudFront substitution recorded in Addendum A. No Phase 0 execution until the Founder lifts the gate after re-verification. Standing rule: no credentials to the AI team or Architecture Team or chat — Bitwarden vault only.